

Development and Experiment Implementation Log: 7 June – 10 June

1. Changes in overall objectives

Between June 7th and June 10th, the project's code implementation goals underwent a significant shift. Initially, the experiments primarily aimed to validate a simple idea: comparing the recommendation performance of traditional collaborative filtering methods with Agentic AI recommendation methods on the H&M e-commerce dataset. However, as issues gradually emerged in the code implementation and early experimental results, the project goal shifted from "creating a demonstrative recommendation system demo" to "building a formal offline recommendation evaluation pipeline that can be used for paper analysis."

In early experiments, several key issues arose with the recommendation and evaluation results. First, some users' ground-truth items were not correctly included in the candidate pool, meaning that even if the model theoretically had a chance to recommend the correct item, a fair comparison could not be formed during the evaluation. Second, `article_id` and `customer_id` inconsistencies in data types exist across different data files or model outputs; for example, some data is processed as integers while others are processed as strings. This can lead to incorrect matching between recommended results and ground-truth items. Third, early CF baselines were too simplistic to serve as strong baseline models in formal papers. Therefore, subsequent work gradually adjusted the baseline to a more formal SVD Matrix Factorisation model and fixed the experimental design to leave-one-out evaluation, a unified candidate pool, Top-10 recommendations, and a bootstrap confidence interval.

Ultimately, the project's formal experimental framework was defined as a comparison of three methods:

1. **SVD Matrix Factorisation baseline:** As a strong baseline for traditional collaborative filtering;
2. **Standalone 3-Agent recommender** Ablation study as a purely agentic recommendation method;
3. **Hybrid SVD + 3-Agent re-ranker:** An enhanced recommendation framework that combines collaborative behavioral signals and agentic evidence reasoning.

Therefore, the research objective is no longer simply to prove whether "Agentic AI is necessarily superior to traditional recommendation systems," but rather to more cautiously analyze whether Agentic AI is better suited as an independent recommendation model in e-commerce recommendations, or as a reranking/augmentation/explainability layer built on a strong collaborative filtering baseline. The results of two 1000-user experiments ultimately support the latter interpretation: the Hybrid framework significantly outperforms Standalone

3-Agent and has the potential to enhance SVD in some ranking quality metrics, but it cannot be interpreted as a complete replacement for SVD.

2. June 8: Clear data and baseline tables were obtained through multiple rounds of Codex Prompts.

The core work on June 8th involved clarifying the data structure, experimental base table, user selection rules, ground-truth definition, and candidate pool construction method through multiple Codex prompts. This step was crucial because all subsequent recommendation models and evaluation metrics rely on this foundational data logic. If the base table or evaluation base table is unstable, even if the subsequent model results can be generated, they cannot be used as credible experimental results for the paper.

A. Data source confirmation

The project uses the H&M Personalized Fashion Recommendations dataset. The formal experiments primarily rely on two core data sources:

1. `transactions_train.csv`

This document provides historical purchase behavior between users and products, serving as a core data source for building collaborative filtering baselines and leave-one-out evaluations. Key fields include:

- `customer_id`
- `article_id`
- `t_that`
- `price`
- `sales_channel_id`

2. `articles.csv`

This document provides product metadata, which is a crucial data source for product evidence analysis, preference matching, and diversity calculations in Standalone 3-Agent and Hybrid reranking. Key fields include:

- `article_id`
- `product_code`
- `prod_name`
- `product_type_name`
- `product_group_name`
- `graphical_appearance_name`
- `colour_group_name`
- `garment_group_name`
- `department_name`
- `section_name`
- `index_name`

- `detail_desc`

`customers.csv` Although it exists, it is not a core modeling field in the current formal experiments. The reason is that the focus of this stage of research is not on user profile attribute modeling, but on comparing the offline ranking performance of SVD, Standalone 3-Agent, and Hybrid reranking based on historical purchase behavior and product metadata.

Therefore, the data basis for the formal experiment was determined as follows:

`transactions_train.csv` + `articles.csv`

Among them, transaction behavior is used to construct user history and SVD collaborative filtering signals, and product metadata is used for agentic preference reasoning, evidence matching, and diversity calculation.

B. Data Base Table Construction

To ensure that different recommendation methods use the same data foundation, a unified processed interaction table needs to be built first. The specific logic is to...`transactions_train.csv` and `articles.csv` pass `article_id` Perform an inner join to create an experimental baseline table containing user purchase behavior and product attribute information.

The core structure of the base table can be understood as:

Each row = a record of a user purchasing a product + the product's metadata

The key fields retained in the base table include:

`customer_id`

`article_id`

`t_dat` / `transaction_date`

`product_type_name`

`product_group_name`

`graphical_appearance_name`

`colour_group_name`

`prod_name`

`detail_desc`

garment_group_name

department_name

section_name

index_name

This base table serves three purposes:

First, it provides the basis for the user-item interaction matrix of the SVD baseline.

Second, it provides Standalone 3-Agent with user history preferences and candidate item evidence.

Third, it provides a unified calculation basis for SVD score, agency score, and diversity bonus for Hybrid reranker.

At this stage, the focus of the code work is not simply reading data, but ensuring that all subsequent models use the same processed dataset to avoid unfair comparisons caused by inconsistent data inputs from different methods.

C. Unified ID type

In the early experiments, a very important question was `article_id` and `customer_id`. The types of inconsistencies are significant. This is an easily overlooked but crucial issue that can severely impact the results of recommender system evaluation.

For example, the ground-truth item might be saved as a string:

"0706016001"

The output of a certain recommendation method `article_id` may be processed as an integer:

706016001

In this situation, even if the model actually recommends the same product, the evaluation code may still judge it as a "miss" because strings and integers cannot be directly matched.

Therefore, the data processing on June 8th explicitly required:

The `article_id` must be treated as a string throughout the entire process.

The `customer_id` must be treated as a string throughout the entire process.

This rule applies to all the following aspects:

- processed interaction table;
- evaluation base table;
- candidate pool;
- SVD recommendation output;
- Standalone 3-Agent recommendation output;
- Hybrid recommendation output;
- metric calculation;
- validation report.

This correction ensures that the calculation of HitRate@10 and NDCG@10 will not produce false negative results due to incorrect ID types.

D. Valid user rules

Formal experiments cannot directly evaluate all users because leave-one-out evaluation requires at least two purchase records for each user: one or more for training history, and the last one as a held-out ground-truth item.

Therefore, the basic rules for valid users are determined as follows:

Users need at least 2 transaction records.

For each valid user:

The last item in the purchase history = ground-truth item

Previous purchase history = training history

If a user has only one transaction record, it is impossible to build both the training history and the ground-truth item at the same time, and therefore the user cannot proceed to the formal evaluation.

This rule ensures that every user being evaluated possesses:

- Historical behavior that can be used for model inference;
- A genuine held-out purchase that can be used for evaluation;
- It is possible to construct ground-truth items for the candidate pool.

This also changed the experimental unit from "transaction records" to "users". In other words, the formal experiment did not evaluate 1,000 transactions, but rather 1,000 valid users.

E. Leave-one-out evaluation rule

The formal experiment employed leave-one-out evaluation. The specific procedure is as follows:

1. Sort the purchase records of each valid user by t_dat ;
2. Treat the last purchase as a held-out ground-truth item;
3. Record previous purchases as training history;
4. Recommendation models can only generate recommendations based on training history;
5. Check if the held-out ground-truth item appears in the Top-10 recommendations.

This design is suitable for offline transaction datasets like H&M's, because the data lacks real impressions, clicks, or online experimental data. By using leave-one-out, an offline recommendation scenario can be simulated: given a user's past purchasing behavior, can the model recommend the product the user will actually buy next from among the candidate products?

This method cannot directly prove improvements in actual user engagement, CTR, or CVR, but it can serve as an evaluation method for offline ranking quality. Therefore, the paper needs to explicitly state that this is an offline evaluation, not a live A/B test.

F. Candidate pool rules

The candidate pool is one of the most critical fairness controls in this phase of the experimental design. Each user's candidate pool is fixed at 100 items.

candidate pool size = 100

It must include:

1 held-out ground-truth item

+ 99 negative candidate items

More importantly, all three methods must use the exact same candidate pool:

SVD, Standalone 3-Agent, and Hybrid use the same candidate pool.

The reason for this is that if different methods deal with different sets of candidate items, the final HitRate@10 , NDCG@10 , and ILD@10 cannot be compared fairly. After unifying the candidate pool, the differences among the three methods mainly come from the ranking logic, rather than the different difficulty of the candidate sets.

During the validation phase, the code needs to be checked:

- Does each user's candidate pool contain a ground-truth item?
- Does the candidate pool contain 100 products?
- Does the candidate pool contain duplicate products?
- Are all of the top 10 recommended candidates from the candidate pool?
- Does the Top-10 recommendation exclude items that the user has already purchased in their training history?
- Do the three methods share the same candidate pool?

Through these rules, the experiment gradually evolved from an unstable early demo into a formal offline evaluation that could be used for paper analysis.

3. Construction of the formal recommendation method

This phase formally retains three recommendation methods for comparison: SVD Matrix Factorisation baseline, Standalone 3-Agent recommender, and Hybrid SVD + 3-Agent reranker. Based on the current focus, this section mainly elaborates on the SVD baseline and the Hybrid framework, while also explaining the role of the Standalone 3-Agent in the experiments.

A. SVD Matrix Factorisation Baseline

SVD Matrix Factorisation was set as the formal collaborative filtering baseline. Its role is to learn the potential relationship structure between users and products based on users' historical purchasing behavior.

Compared to earlier, simpler CF methods, SVD is more suitable as a formal baseline for the following reasons:

First, SVD is a common matrix factorization method in recommendation systems, which can map sparse user-item interaction matrices to low-dimensional latent factor spaces.

Second, it is based on the interaction between users and products, rather than relying solely on product attributes, thus enabling it to capture collaborative purchasing patterns.

Third, it can serve as a relatively strong traditional recommendation model to determine whether agentic AI methods truly bring incremental value.

The basic input for SVD is:

user-item interaction matrix

Among them, users are `customer_id` The product is `article_id` The interaction signals come from the user's historical purchase records. After the model is trained, it calculates recommendation scores for users and candidate items, and then sorts them among each user's 100 candidate items, outputting a Top-10 recommendation.

In the experimental design, the recommendation results of SVD are not freely output from the entire product database, but are restricted to the same candidate pool:

SVD ranking = Sort and output the Top-10 from the user's 100 candidate items.

This ensures a fair comparison between SVD, Standalone 3-Agent, and Hybrid.

Experimental results show that SVD significantly outperforms Standalone 3-Agent in both 1000-user evaluations. This indicates that in the H&M dataset, the collaborative signal from users' historical purchase behavior is crucial for offline ranking performance. This finding also provides the basis for the paper's conclusion: the agentic method should not be simply described as a replacement for traditional collaborative filtering, but rather as a suitable enhancement layer.

B. Experimental use of Standalone 3-Agent Recommender

Although the user's current focus is not on expanding on Standalone 3-Agent, its role still needs to be explained in the overall experiment. Standalone 3-Agent is an ablation method used to test whether pure agentic preference reasoning is strong enough when it does not rely on SVD behavior scores.

Standalone 3-Agent mainly includes:

Preference Agent

Evidence / Item Agent

Decision Agent

Its basic logic is:

- The Preference Agent extracts preferences from the user's training history;
- Evidence / Item Agent reads product attribute evidence from candidate items;
- The Decision Agent ranks candidate products based on user preferences and product evidence.

This method does not use SVD score, and therefore can serve as an independent control for the agentic component in the hybrid method. Two 1000-user experiments showed that the standalone 3-Agent was significantly weaker than SVD, indicating that relying solely on product metadata and agentic reasoning cannot replace collaborative filtering of behavioral signals.

This result should not be hidden. On the contrary, it is an important finding in the paper: Agentic AI has limited effectiveness in ranking on its own, but when combined with SVD, it can perform better as a reranking and explanation layer.

C. Hybrid SVD + 3-Agent Reranker

Hybrid SVD + 3-Agent reranker is the most important recommendation framework. It does not completely replace SVD, but adds agentic scoring and diversity bonus to SVD to re-rank candidate items.

The fixed formula for Hybrid is:

hybrid_score =

0.70 * normalized_svd_score

+ 0.25 * normalized_agentic_score

+ 0.05 * diversity_bonus

in:

normalized_svd_score

This represents the behavioral recommendation score of SVD for candidate products, primarily capturing the collaborative filtering signal between users and products.

normalized_agentic_score

This represents the matching score calculated by the agentic component based on user preferences and product metadata evidence, mainly reflecting the interpretive matching of product attributes, color, appearance, category, and historical preferences.

diversity_bonus

To alleviate the over-concentration of the recommendation list, we attempted to incorporate some diversity considerations into the sorting process.

Hybrid's design logic is as follows: SVD provides the main sorting capabilities, the Agentic component provides preference interpretation and product evidence matching, and the diversity bonus attempts to avoid the recommendation list from being too simplistic.

Based on the results of two 1000-user experiments, Hybrid significantly outperformed Standalone 3-Agent, and in the second seed99 experiment, NDCG@10 significantly outperformed SVD. This suggests that Hybrid may have incremental value in "ranking the right products higher." However, Hybrid did not consistently and significantly outperform SVD

at HitRate@10, and ILD@10 was lower than SVD in both experiments. Therefore, Hybrid's final positioning should be:

augmentation / reranking / explainability layer

Instead of:

full replacement for SVD

4. Indicator System Organization

The formal experiment used three core offline ranking metrics:

Metric	Chinese explanation	Role in this experiment
HitRate@10	Did the Top-10 hit the ground-truth item?	The measure is whether the model can include users' actual next purchases in the Top-10.
NDCG@10	Quality of the position of the hit product in the Top-10	The measure is whether the model can prioritize ground-truth items.
ILD@10	Intra-list Diversity	Measuring the diversity among products within a recommendation list

HitRate@10

The calculation logic for HitRate@10 is as follows:

If a held-out ground-truth item appears in the Top-10 recommended results, then the user's Hit = 1;

Otherwise, Hit = 0.

Ultimately, HitRate@10 is the average hit rate for all valid users.

This metric can be interpreted as a fundamental indicator of offline recommendation relevance, but it cannot be directly interpreted as CTR or conversion rate. This is because the H&M dataset lacks impression or click data, meaning the model has not been tested online by real users.

NDCG@10

NDCG@10 considers not only whether the ground-truth item appears in the Top-10, but also its position within the Top-10. A higher score is given if the ground-truth item is ranked #1, and a lower score if it's ranked #10.

Therefore, NDCG@10 is more suitable for measuring ranking position quality.

In the seed99 experiment, the difference in NDCG@10 between Hybrid and SVD is:

[0.007190, 0.025000]

The fact that this interval does not cross 0 indicates that in the second 1000-user robustness run, Hybrid significantly outperforms SVD in terms of sorting position quality. However, this still does not directly prove that Hybrid is superior to SVD across the board, because the difference CI for HitRate@10 crosses 0.

ILD@10

ILD@10 measures the diversity within the recommendation list. A higher value indicates greater differences between products in the Top-10 recommendation list; a lower value indicates a more concentrated recommendation result.

In both 1000-user experiments, SVD's ILD@10 was significantly higher than Hybrid's, indicating that while Hybrid improves ranking quality to some extent, it sacrifices some diversity.

Therefore, the following cannot be written in the final paper:

Hybrid improves diversity.

A more accurate way to write it is:

Hybrid improves over standalone 3-agent and remains competitive with SVD on ranking quality, but it consistently reduces intra-list diversity compared with SVD.

Why not just use the CTR/CVR/CE metrics?

This project uses an offline transaction dataset, which does not include the following online behavioral data:

impression

click

dwel time

add-to-cart

conversion funnel

explicit feedback

online A/B test result

Therefore, the experiment cannot directly prove that real customer engagement, CTR, or CVR are improved. HitRate@10 and NDCG@10 can at most serve as offline proxies for recommendation relevance/ranking quality. This limitation must be explicitly stated in the paper.

5. First formal 1000-user experiment

The first formal 1000-user experiment used 1000 valid users, each undergoing leave-one-out evaluation, and a 100-item candidate pool. All three methods used the same candidate set, outputting Top-10 recommendations and calculating HitRate@10, NDCG@10, and ILD@10.

The first official results are as follows:

Method	HitRate@10	NDCG@10	ILD@10
SVD	0.500000	0.295493	0.751933
Standalone 3-Agent	0.274000	0.155365	0.543800

Hybrid	0.498000	0.304552	0.694481
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The first experiment shows that SVD significantly outperforms Standalone 3-Agent on HitRate@10 and NDCG@10. This indicates that the pure agentic recommendation method, when used alone as a ranking model, cannot effectively replace the collaborative filtering baseline.

Hybrid shows a significant improvement over Standalone 3-Agent. Hybrid's HitRate@10 is 0.498, close to SVD's 0.500, and its NDCG@10 is 0.304552, higher than SVD's 0.295493. This suggests that Hybrid may offer incremental value in terms of sorting position quality.

However, in the first experiment, Hybrid's ILD@10 was 0.694481, lower than SVD's 0.751933, indicating that Hybrid's recommendations were more concentrated and list diversity was reduced. Therefore, the first experiment cannot support the conclusion that "Hybrid is comprehensively superior to SVD" or "Hybrid improves diversity."

The safety explanation for the first experiment is:

SVD is a strong baseline.

Standalone 3-Agent is weak as an independent recommender.

Hybrid strongly improves over Standalone 3-Agent and remains competitive with SVD.

However, Hybrid does not clearly replace SVD and reduces intra-list diversity.

6. Seed99 Second 1000-user robustness experiment

To verify the stability of the first 1000-user results, the project ran a second 1000-user robustness experiment with a random seed of 99. The second experiment maintained the same evaluation protocol, candidate pool size, Top-K metrics, and hybrid formula, only changing the random seed and user sampling.

The results of the second seed99 test are as follows:

Method	HitRate@10	NDCG@10	ILD@10
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SVD	0.505000	0.296245	0.755089
Standalone 3-Agent	0.269000	0.154352	0.539319
Hybrid	0.526000	0.312443	0.692919

The second experiment again showed that SVD significantly outperformed Standalone 3-Agent:

SVD HitRate@10 = 0.505000

Standalone 3-Agent HitRate@10 = 0.269000

This is completely consistent with the first experiment, further illustrating the importance of collaborative filtering behavior signals in the H&M dataset.

Hybrid also clearly outperformed Standalone 3-Agent once again:

Hybrid HitRate@10 = 0.526000

Standalone 3-Agent HitRate@10 = 0.269000

Bootstrap pairwise difference also supports this judgment:

Hybrid minus Standalone 3-Agent HitRate@10 CI = [0.225950, 0.286000]

Hybrid minus Standalone 3-Agent NDCG@10 CI = [0.133920, 0.180924]

Hybrid minus Standalone 3-Agent ILD@10 CI = [0.146976, 0.160104]

None of these intervals cross 0, indicating that Hybrid significantly outperforms Standalone 3-Agent in seed99.

Comparisons between Hybrid and SVD require more caution. In seed99, Hybrid's HitRate@10 is higher than SVD's.

Hybrid HitRate@10 = 0.526000

SVD HitRate@10 = 0.505000

However, the HitRate@10 CI for Hybrid minus SVD is:

[-0.001000, 0.042000]

The interval crosses 0, so it cannot be said that Hybrid significantly outperforms SVD on HitRate@10.

Meanwhile, Hybrid's NDCG@10 is higher than SVD's:

Hybrid NDCG@10 = 0.312443

SVD NDCG@10 = 0.296245

The NDCG@10 CI of the Hybrid minus SVD is:

[0.007190, 0.025000]

The fact that this interval does not cross 0 indicates that Hybrid in seed99 is significantly better than SVD in terms of ranking position quality.

However, the Hybrid's ILD@10 is still lower than SVD:

Hybrid ILD@10 = 0.692919

SVD ILD@10 = 0.755089

The ILD@10 CI for the Hybrid minus SVD is:

[-0.066816, -0.057555]

This interval is entirely below 0, indicating that the recommendation diversity for Hybrid is significantly lower than that for SVD.

Therefore, seed99's safety conclusion is:

Hybrid shows stronger NDCG@10 than SVD in the second 1,000-user sample, but its HitRate@10 advantage is not statistically clear and its diversity remains lower. Therefore, Hybrid should still be framed as a reranking and augmentation layer rather than a full replacement for SVD.

7. Are the two 1000-user experiments consistent?

The values from the two 1000-user experiments are not exactly the same, but the principal direction height is consistent. The comparison is as follows:

Method	Run 1 HitRate@10	Seed99 HitRate@10	Run 1 NDCG@10	Seed99 NDCG@10	Run 1 ILD@10	Seed99 ILD@10
SVD	0.500000	0.505000	0.295493	0.296245	0.751933	0.755089
Standalone 3-Agent	0.274000	0.269000	0.155365	0.154352	0.543800	0.539319
Hybrid	0.498000	0.526000	0.304552	0.312443	0.694481	0.692919

As can be seen from the table, SVD performed very stably in both experiments. HitRate@10 increased from 0.500 to 0.505, NDCG@10 increased from 0.295493 to 0.296245, and ILD@10 increased from 0.751933 to 0.755089, with very little change.

The Standalone 3-Agent also showed consistently weak performance in both experiments. HitRate@10 decreased from 0.274 to 0.269, NDCG@10 decreased from 0.155365 to 0.154352, and ILD@10 decreased from 0.543800 to 0.539319. This indicates that its weak performance in the first experiment was not accidental, but rather that it continued to lag significantly behind in the second independent sampling.

Hybrid performed slightly better in the second experiment than in the first. HitRate@10 increased from 0.498 to 0.526, and NDCG@10 increased from 0.304552 to 0.312443, but ILD@10 remained relatively similar, decreasing from 0.694481 to 0.692919. This indicates that Hybrid had stronger ranking quality in the second batch of users, but the diversity trade-off still exists.

The consistent conclusions from both experiments include:

1. **SVD consistently outperforms Standalone 3-Agent.**
This indicates that user behavior coordination signals remain the core signals in e-commerce recommendations.
2. **Hybrid consistently outperforms Standalone 3-Agent.**
This shows that the agentic method is weak when used alone, but is significantly more effective when combined with SVD.
3. **Hybrid remains competitive with SVD on ranking quality.**
In the first experiment, the Hybrid's HitRate@10 was close to SVD, while its NDCG@10 was slightly higher; in the second experiment, the Hybrid's NDCG@10 was significantly higher than SVD, but the advantage of HitRate@10 was not significant.
4. **Hybrid consistently has lower diversity than SVD.**
In both experiments, the ILD@10 of Hybrid was lower than that of SVD, indicating that the recommended results were more concentrated after reranking.
5. **Hybrid should not be framed as a full replacement for SVD.**
Because it did not consistently and significantly outperform SVD across all metrics, especially HitRate@10 and ILD@10.

Therefore, the final judgment is:

The two 1,000-user experiments are directionally consistent. They do not show that Hybrid universally replaces SVD, but they do support the use of Hybrid SVD + agentic reranking as an augmentation layer that can improve ranking position quality and provide evidence-based explainability while maintaining competitive offline relevance.

8. Finally, it can be used for the concluding statement of the paper.

Two 1000-user offline evaluations together support the following conclusion:

Across two independent 1,000-user offline evaluations, the SVD baseline consistently outperformed the standalone 3-agent recommender, confirming that behavioural collaborative filtering signals remain important in the H&M e-commerce recommendation setting. The Hybrid SVD + agentic reranking framework consistently and substantially outperformed the standalone 3-agent method, showing that the agentic component is more effective when used as an augmentation and reranking layer rather than as an independent recommender. Compared with SVD, the Hybrid method remained competitive in HitRate@10 and achieved stronger NDCG@10 in the seed99 robustness run, suggesting improved ranking position quality. However, the Hybrid method did not demonstrate a consistently significant HitRate@10 advantage over SVD and consistently reduced intra-list diversity. Therefore, the Hybrid framework should be interpreted as a reranking and explainability-oriented augmentation of SVD, rather than a complete replacement.